



OpenMined

AI ALLIANCE



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**Data infrastructure for frontier
sovereign AI**



OpenMined

OpenMined is a 501(c)(3) non-profit foundation and mission-driven community that brings together leading minds in AI, security and privacy technology to develop open-source solutions that transform how data can be accessed and shared.

AISI AI SAFETY
INSTITUTE

reddit

 Microsoft

DAILYMOTION

DeepMind



 Meta

Google

 UNIVERSITY OF
OXFORD

 GOV.UK

 UN PET Lab
United Nations
Privacy Enhancing Technologies Lab

BASELINE ASSUMPTION:

For sovereign AI to be successful, it needs
to *outperform* private frontier models

Proposition: access to rare data is a winning strategy

Good AI needs good data.

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Every big leap in AI came from more good data.



Artificial intelligence (AI)

This article is more than 4 months old

Elon Musk says all human data for AI training 'exhausted'

Tech boss suggests move to self-learning synthetic data though some warn this could cause 'model collapse'

Business live - latest updates

Dan Milmo
Global technology editor

Thu 9 Jan 2025 12:14 EST

Share



Elon Musk: 'The cumulative sum of human knowledge has been exhausted in AI training. That happened basically last year.' Photograph: Allison Robbert/Reuters

Artificial intelligence companies have run out of data for training their models and have "exhausted" the sum of human knowledge, **Elon Musk** has said.

Sourcegraph

AI-Driven Enterprise Security.

Learn more

BUSINESS • TECHNOLOGY

Ilya Sutskever Warns A.I. Is Running Out of Data—Here's What Will Happen Next

"We've achieved peak data and there will be no more," said the OpenAI co-founder.

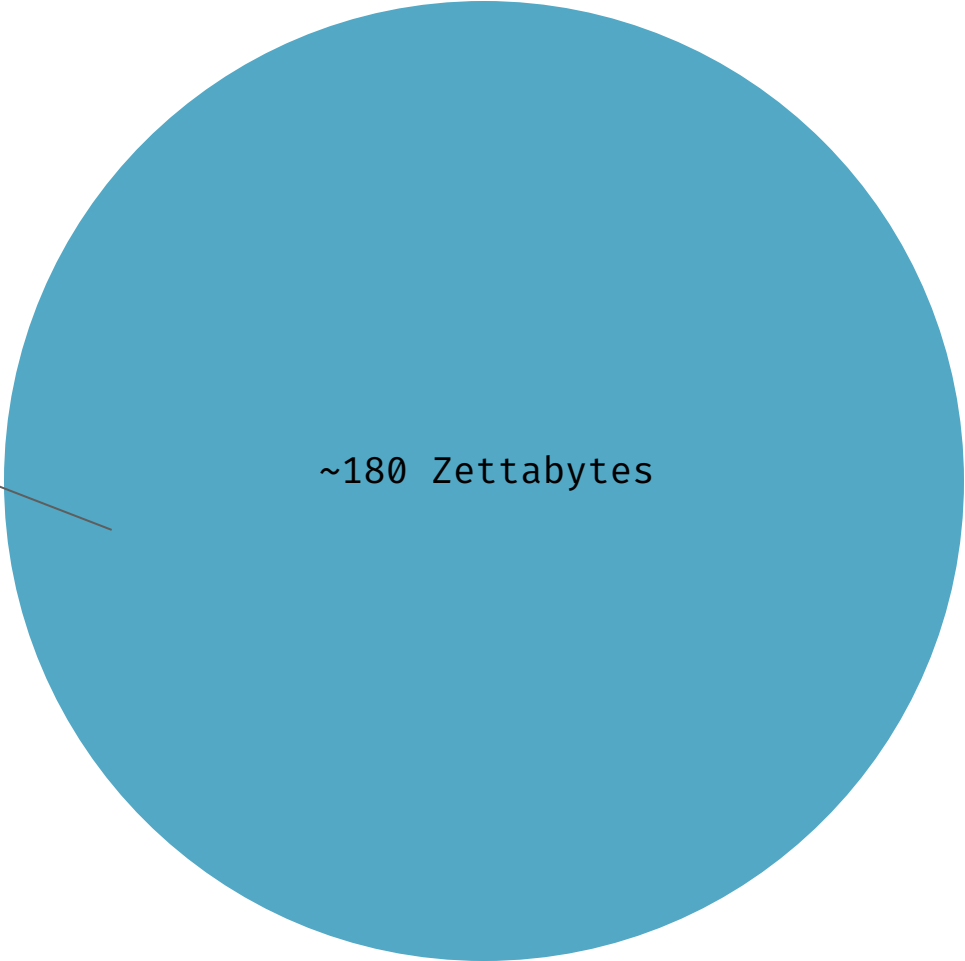
By Alexandra Tremayne-Pengelly • 12/16/24 4:08pm

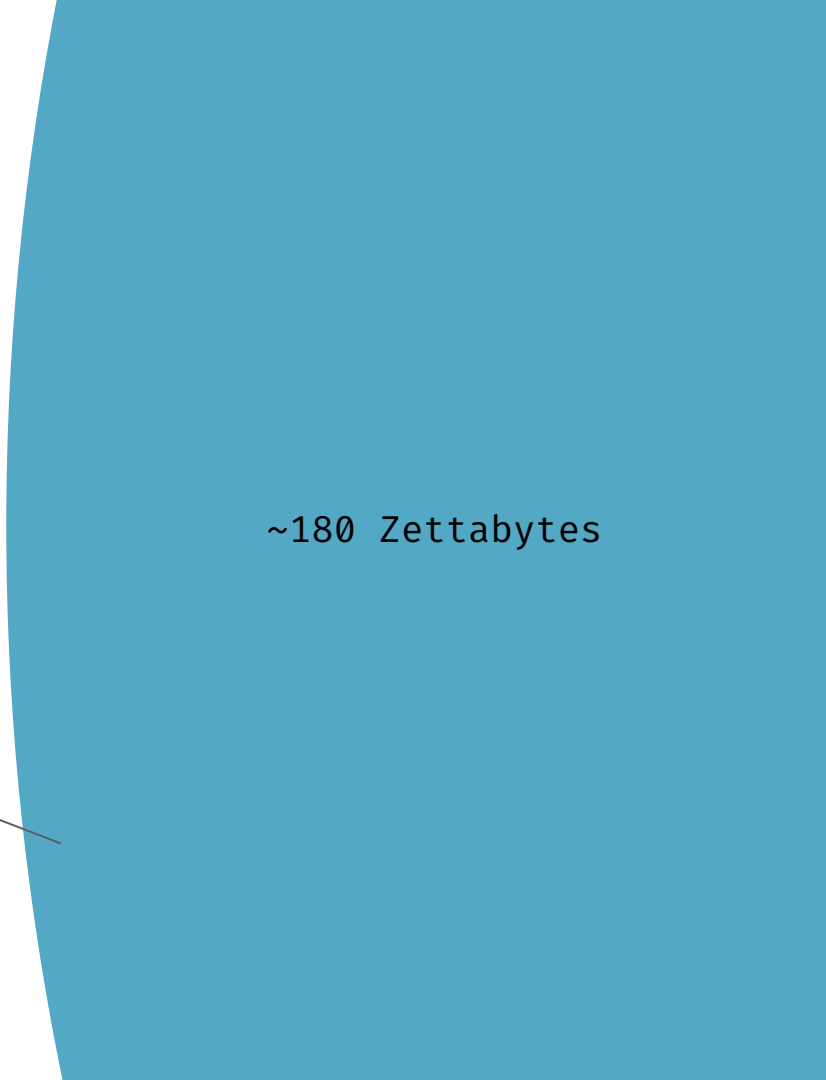


Ilya Sutskever founded the A.I. startup Safe Superintelligence after leaving OpenAI. AFP via Getty Images

All the data that exists

~180 Zettabytes

A large, solid blue circle is centered on the page. It represents the total amount of data that exists. A thin black line extends from the left edge of the circle towards the text 'All the data that exists'.

A large, solid blue curved shape that starts from the right edge of the frame and curves towards the left, defining a large area. It represents a significant volume of data.

~180 Zettabytes

All the data that exists

A thin black line originates from the text 'All the data that exists' and points diagonally upwards and to the right, ending at the boundary of the blue curved shape.

Largest AI training dataset

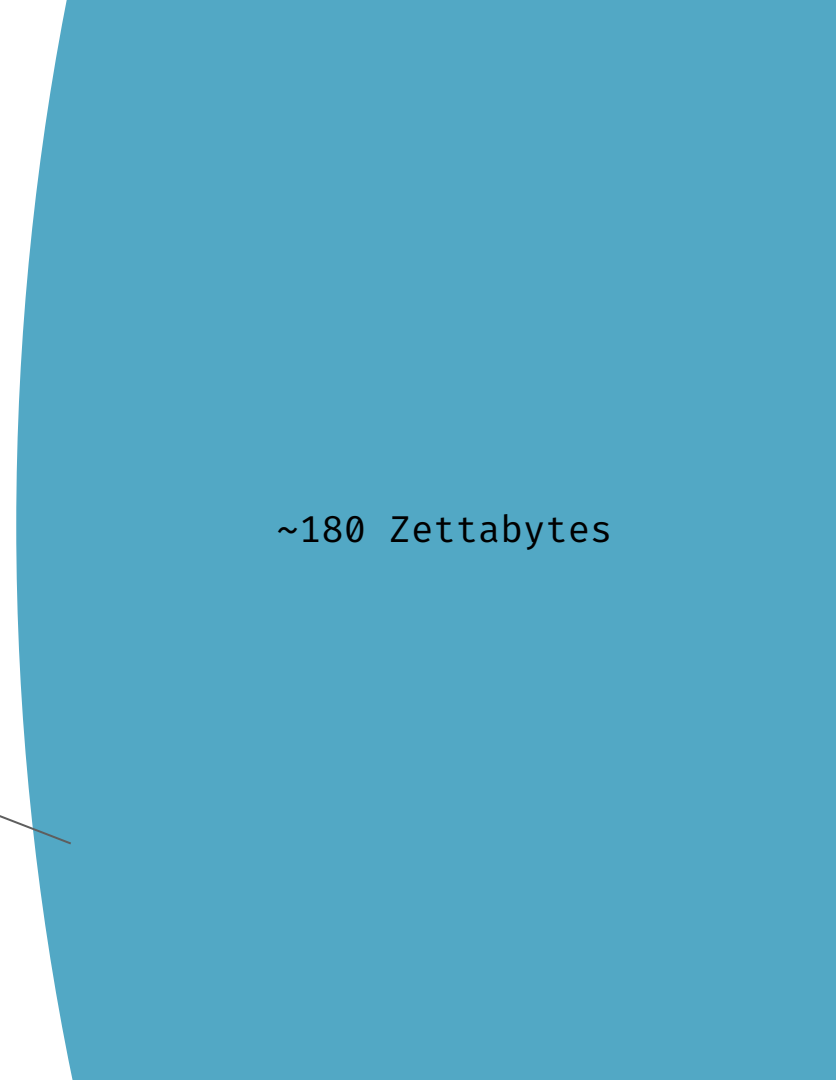


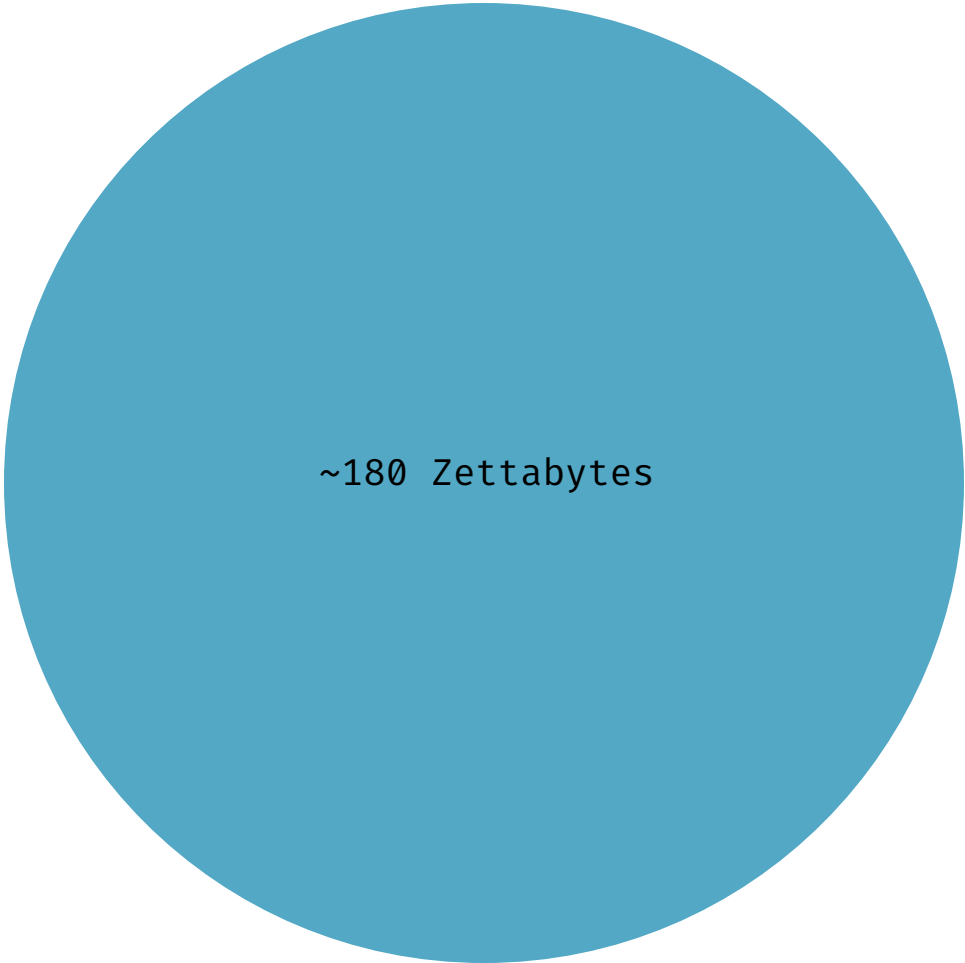
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All the data that exists

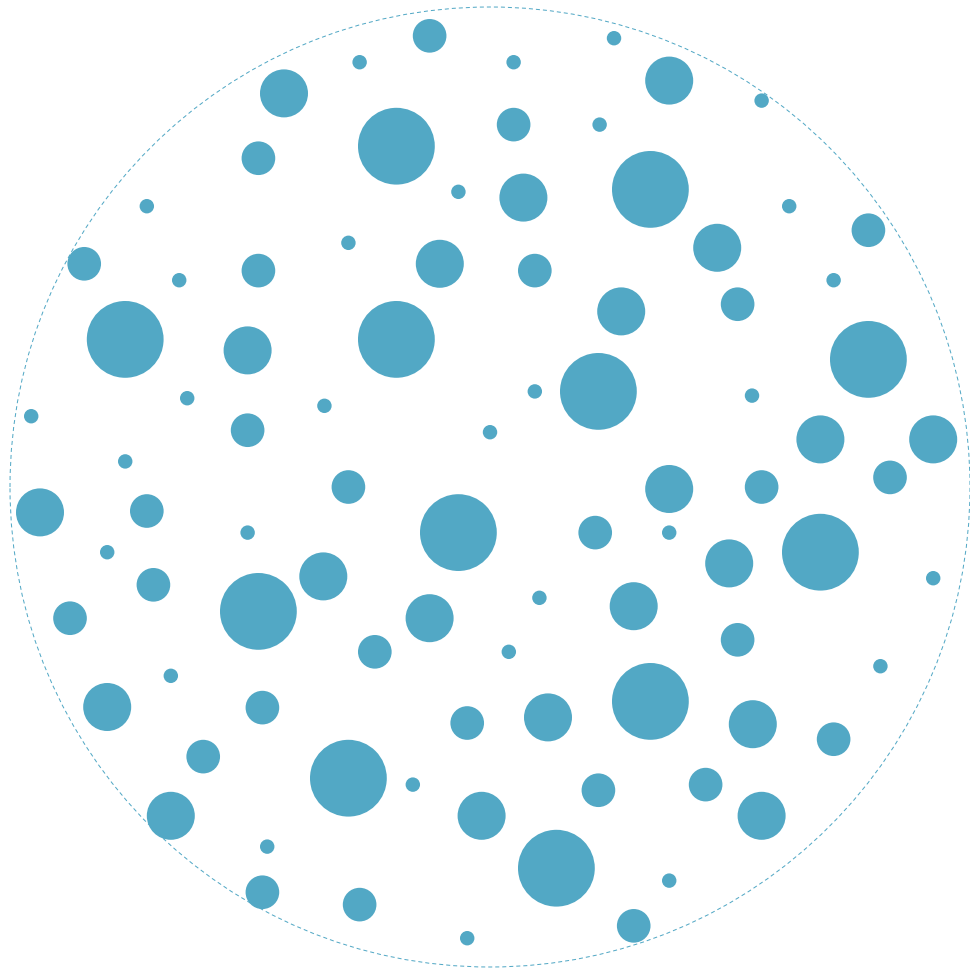


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~180 Zettabytes



**The next frontier of AI is data that is
not on the public web...**

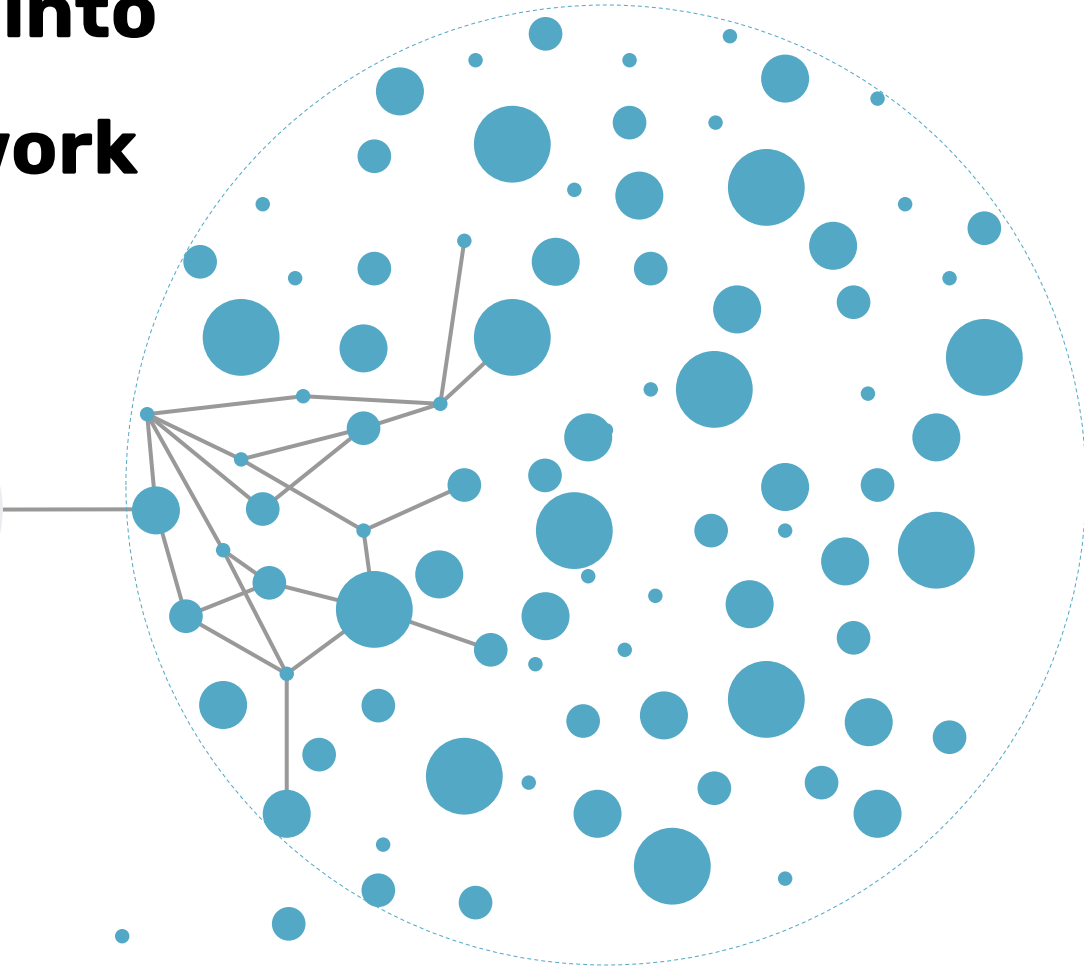
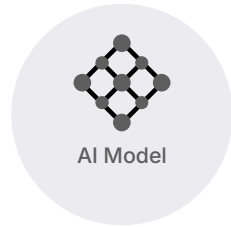
**The next frontier of AI is data that is
not on the public web...**

**...and it shouldn't be handed to a
centralised provider**

THE CHALLENGE:

**Can we build public-interest,
frontier-level AI whilst retaining data
sovereignty?**

**What if we could tap into
a decentralised network
of data holders?**



POSSIBLE SOLUTION:

Network-Sourced AI

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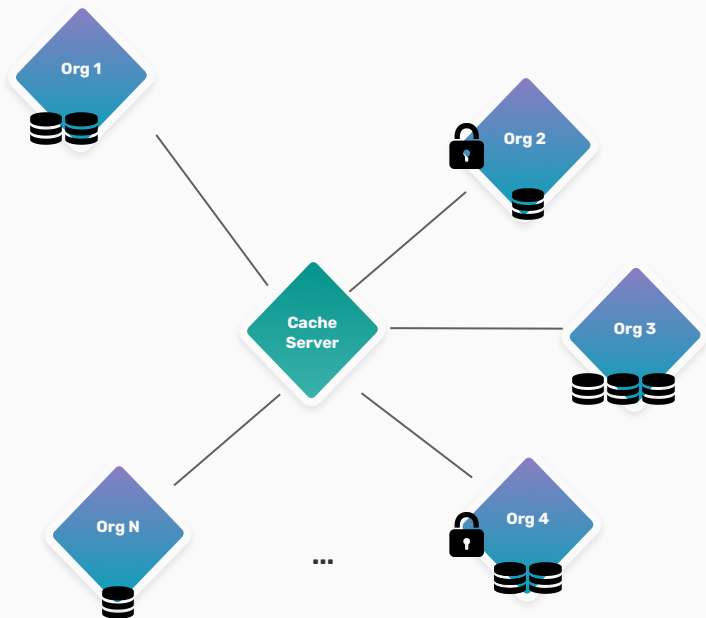
Network-Sourced AI

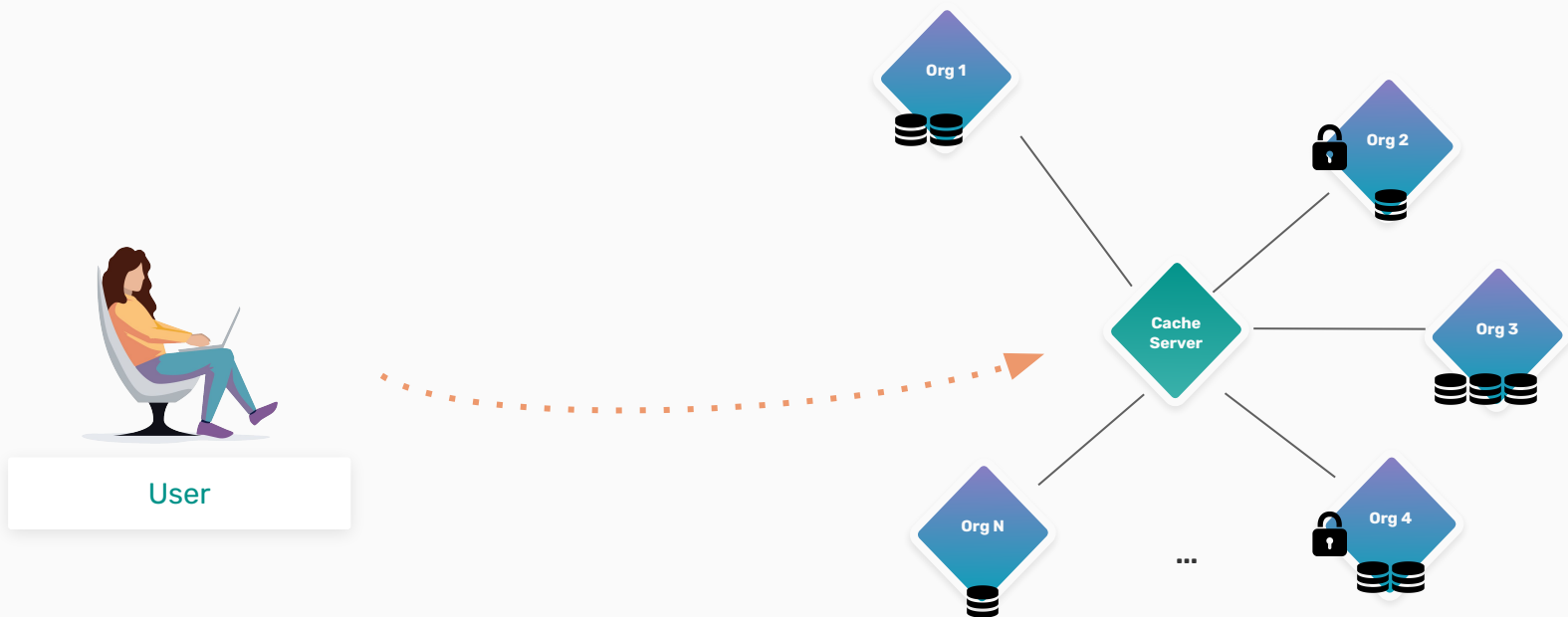
Data owners: *data sovereignty via attribution-based control*

AI users: *open AI capability*



User





1

Discover data on the network

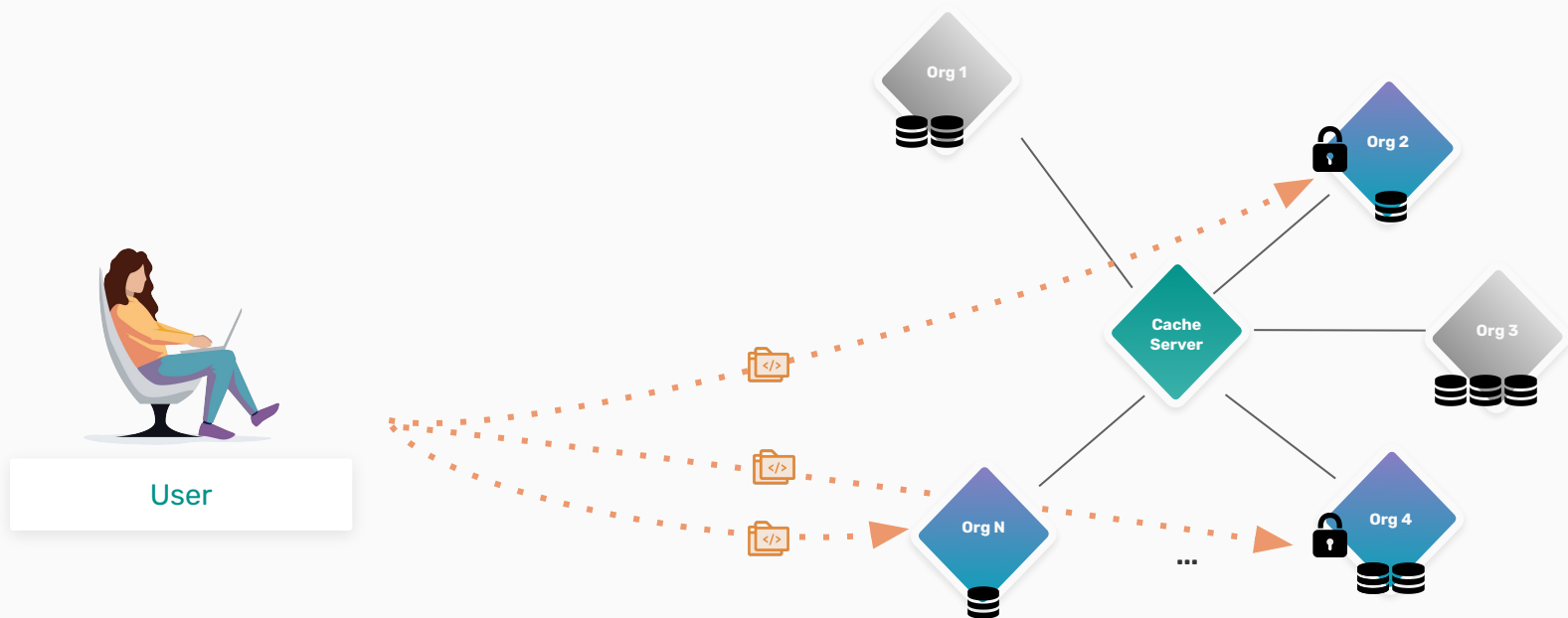
2

Request usage

3

Combine inputs for better predictions





1

Discover data on
the network

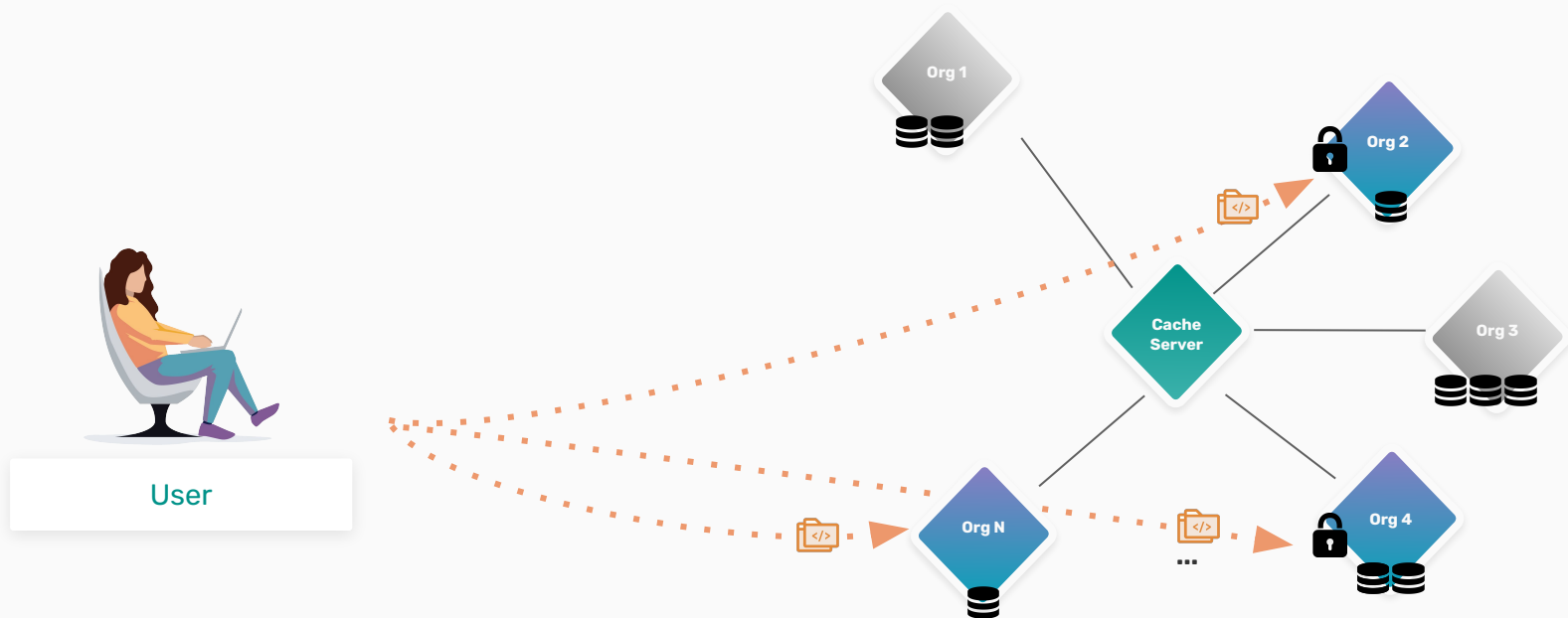
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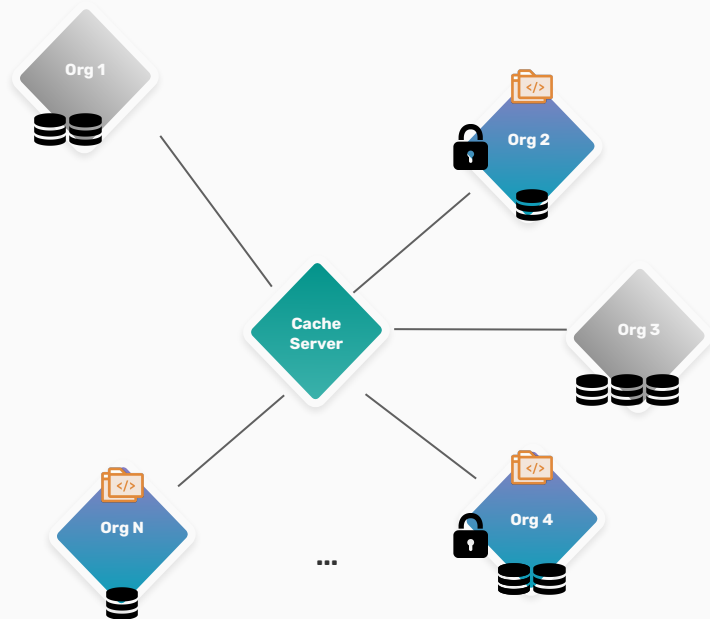
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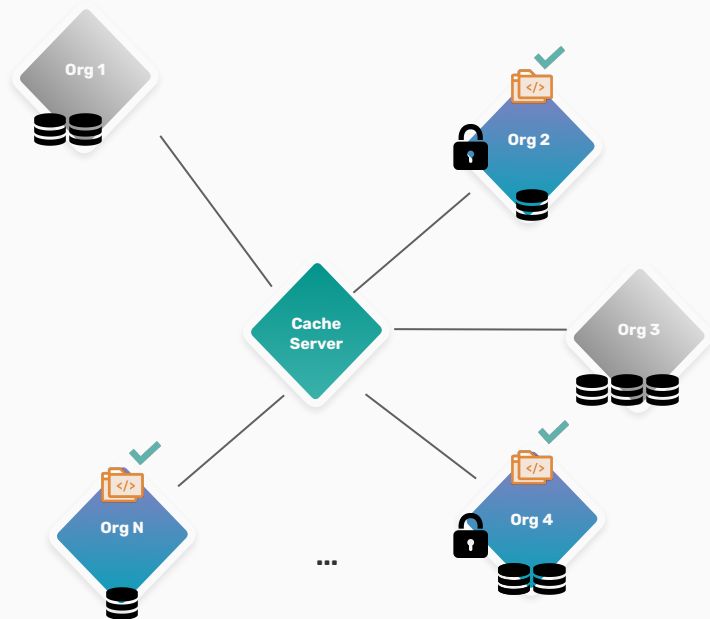
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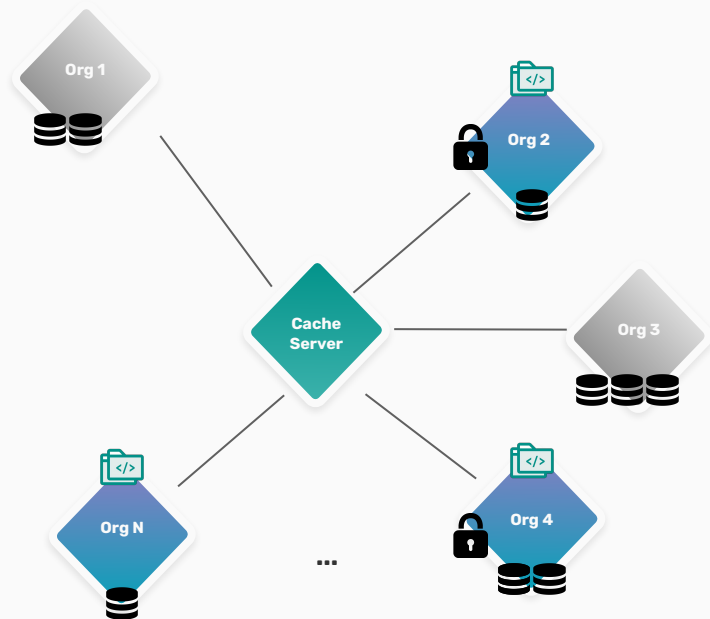
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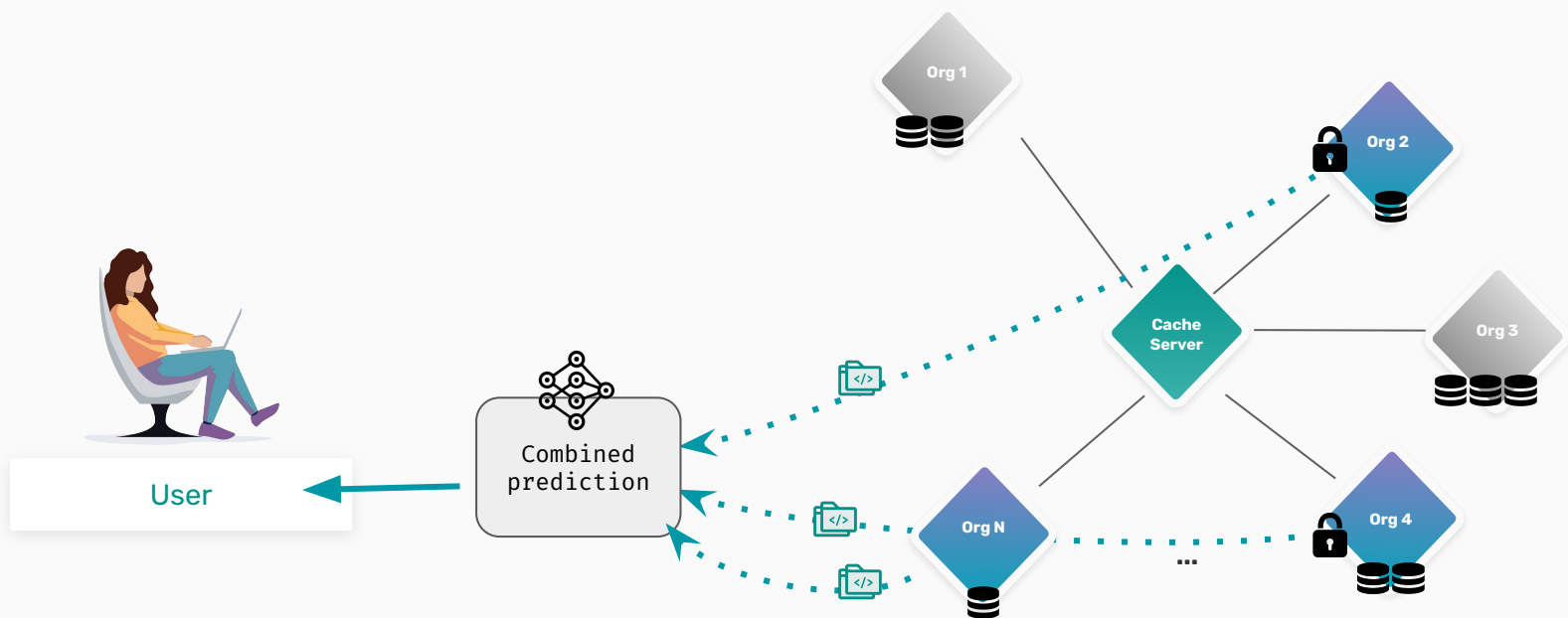
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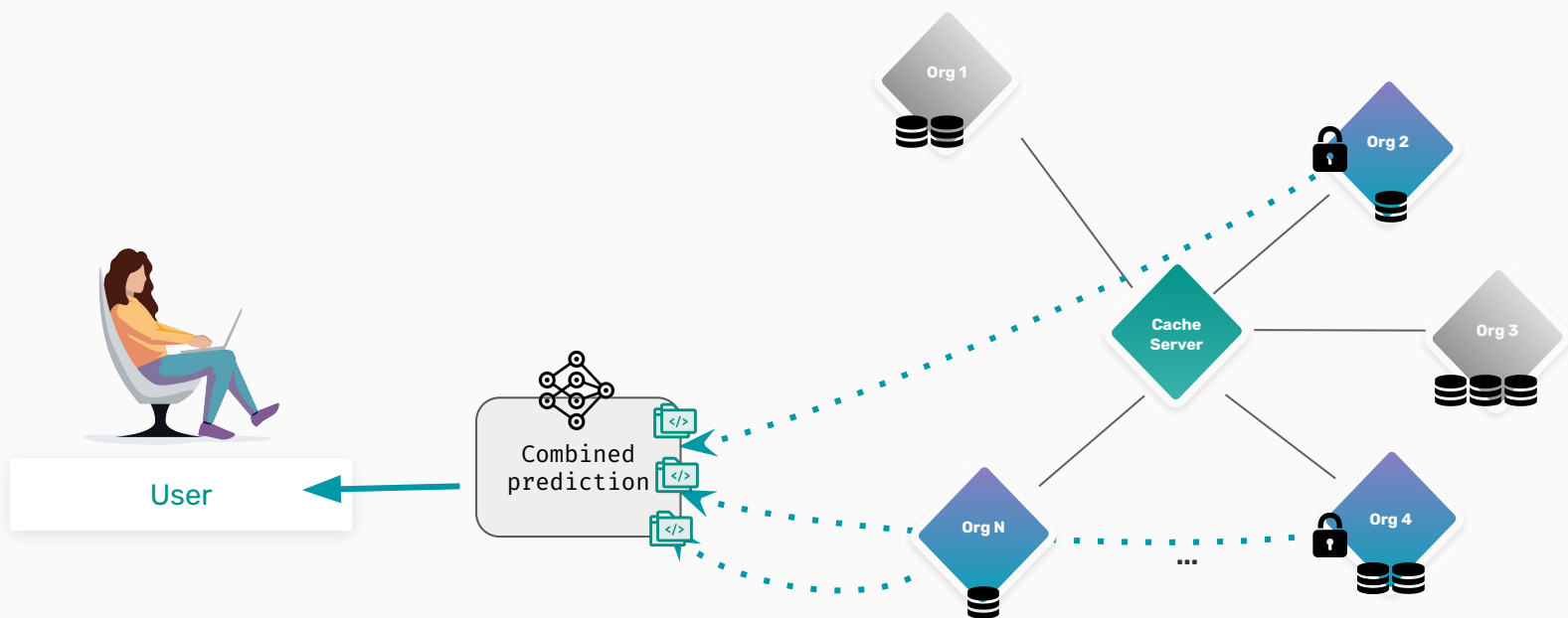
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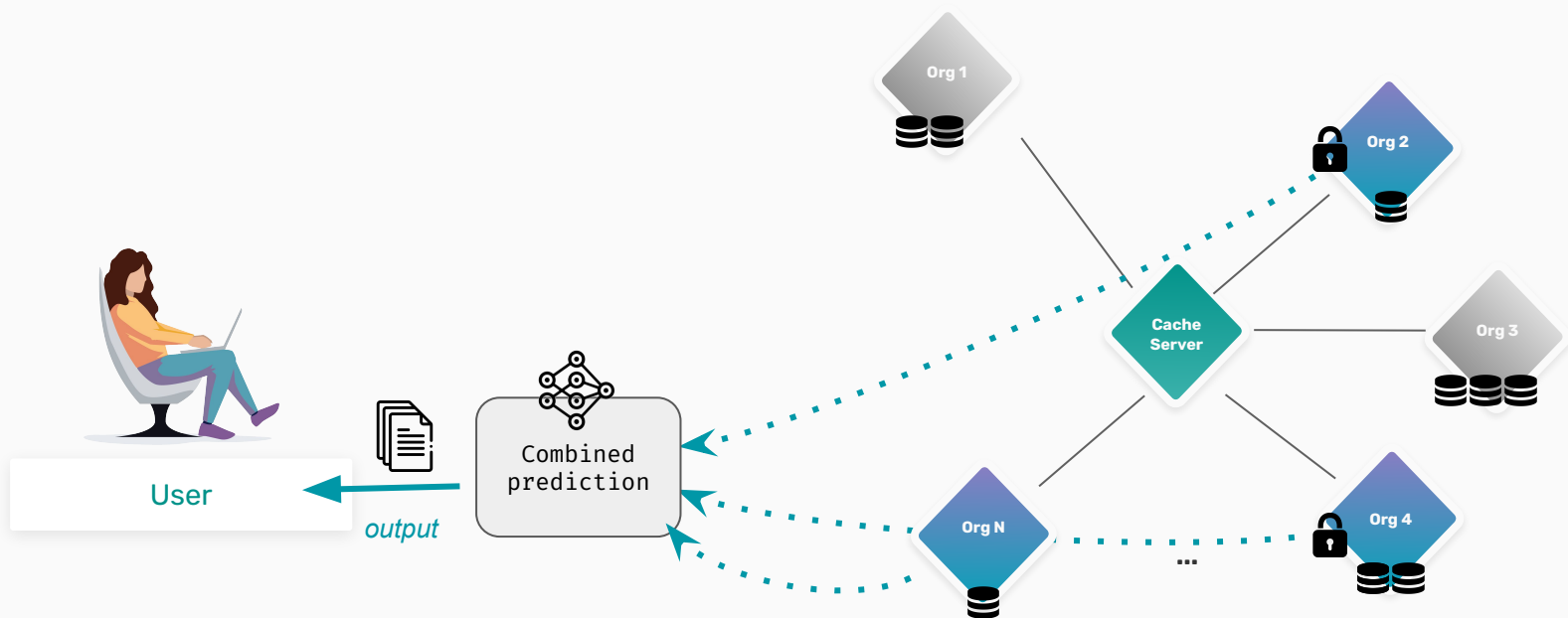
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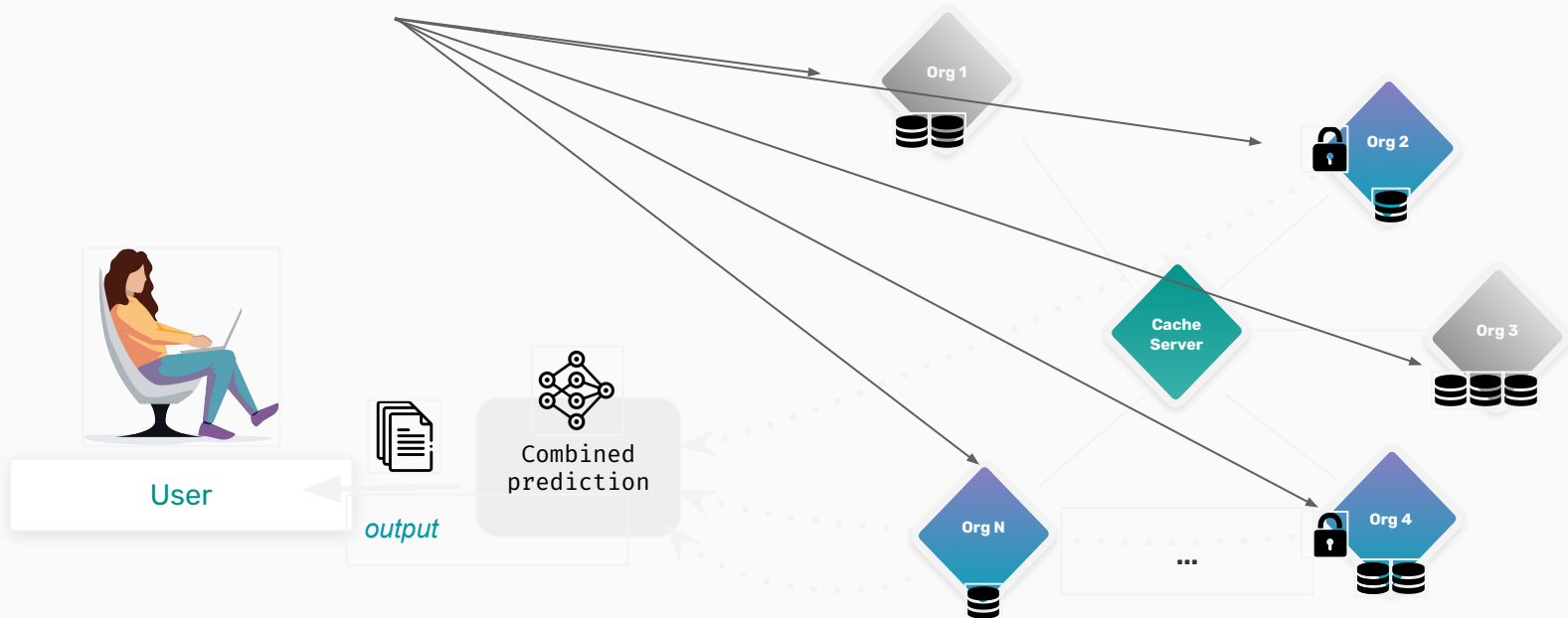
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Closed Source AI Assets



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Discover data on
the network

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Request usage

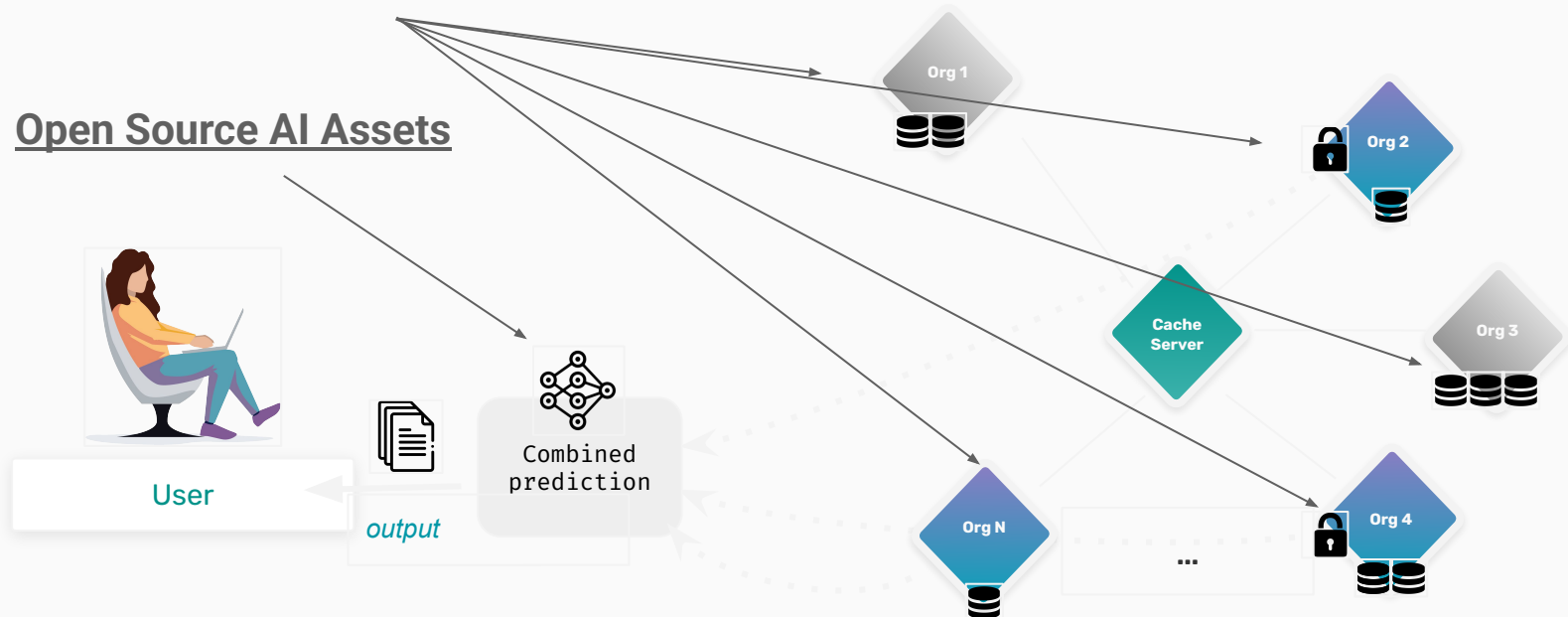
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Closed Source AI Assets

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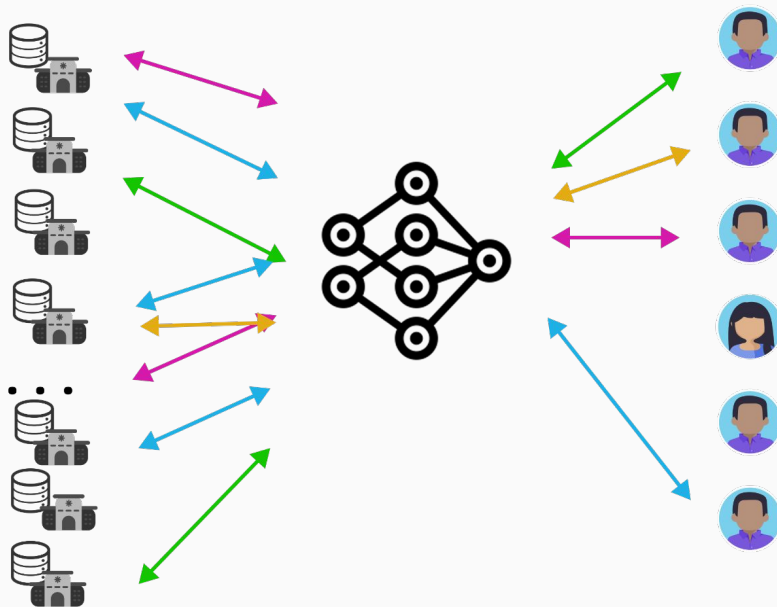
Request usage

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This is “Network-source AI”



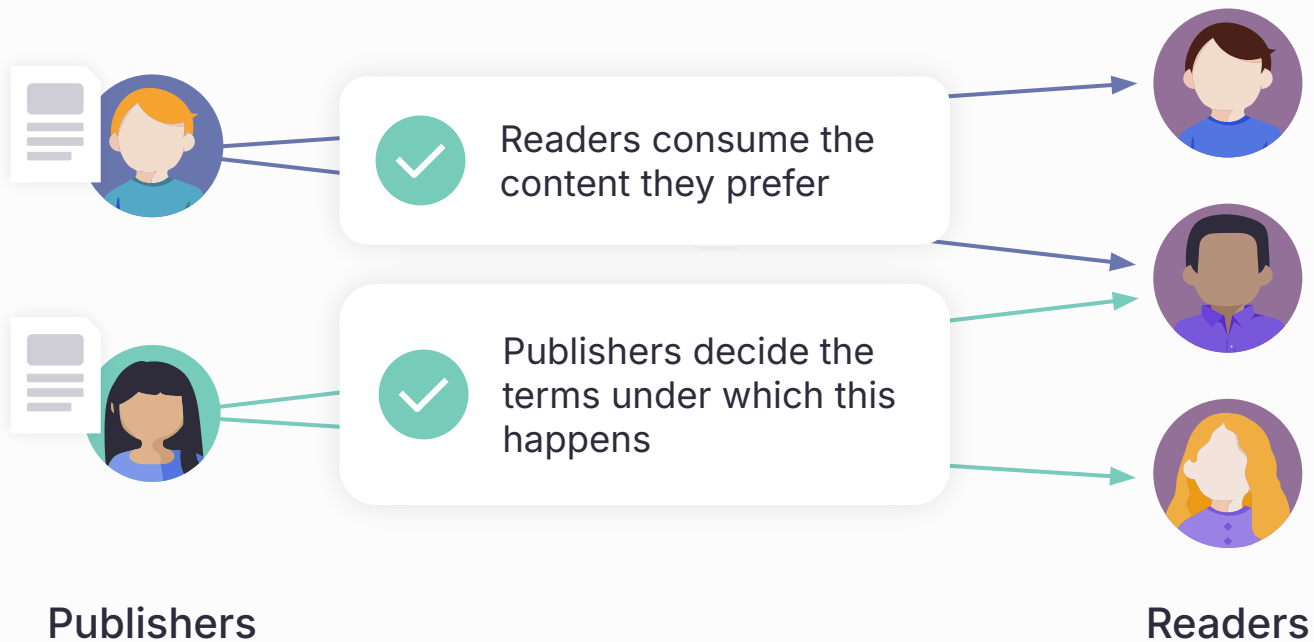
DATA PROVIDERS
with insights

AI USERS
*looking for
insights*

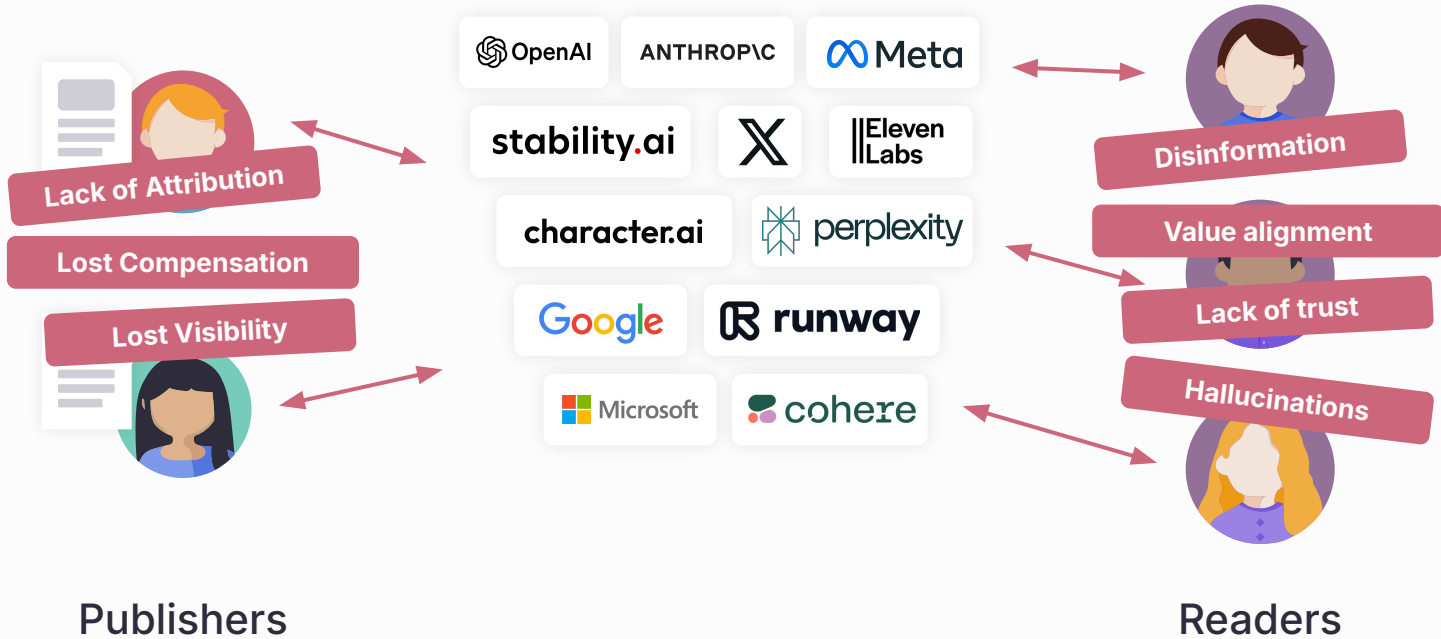


Real-world example: AI that can draw on
high-quality, local journalism

How it Used to Work



What AI Broke



Why publishers lose leverage and credit in AI systems?



Once it's copied, it's gone

Hand over a copy of your content and you have no say in how it is being used anymore.

You're handing over the means to build products that can compete directly with you.



Once it's blended in, you can't trace it

When AI learns from your work, it gets mixed with millions of other sources. There's no technical way to point back and say "that came from us."

What looks like memorization (and give some hope) is an accident due to overtraining, not a mechanism that preserves origin.

Why publishers use leverage and credit in AI systems



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Both copy & distillation problems are mechanisms by which control transfers from publishers to platforms.

Why you cannot fix it “after the fact”



Tracing it back

Trying to work out which source shaped an answer after training is an active research area and falls apart at scale.



Once it's blended in, you can't delete it

“Removing” your content from a trained model rarely sticks, and it makes the model worse in the process.

Why you cannot fix it “after the fact”



Tracing it back

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Once it's blended in, you can't delete it

“Removing” your content from a trained model rarely sticks, and it makes the model worse in the process.

The answer: don't bake work into the model. Let the AI query content directly, at inference time, so it can be directly attributed.

AI with attribution-based control

The old way

- Your work gets blended in and none can see where the answer came from
- It's copied, so you lose control

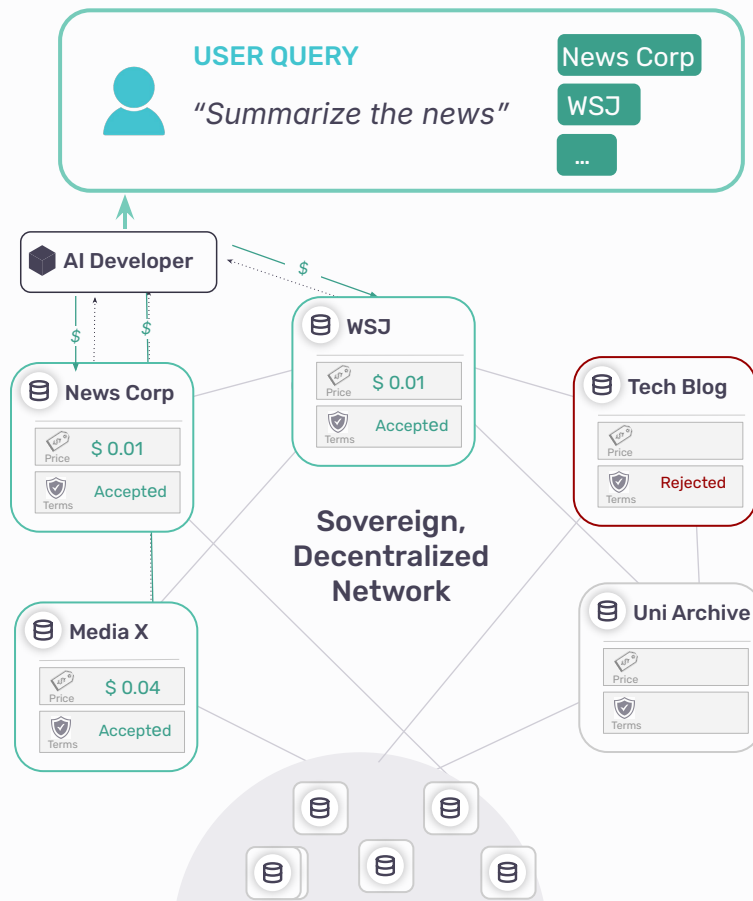
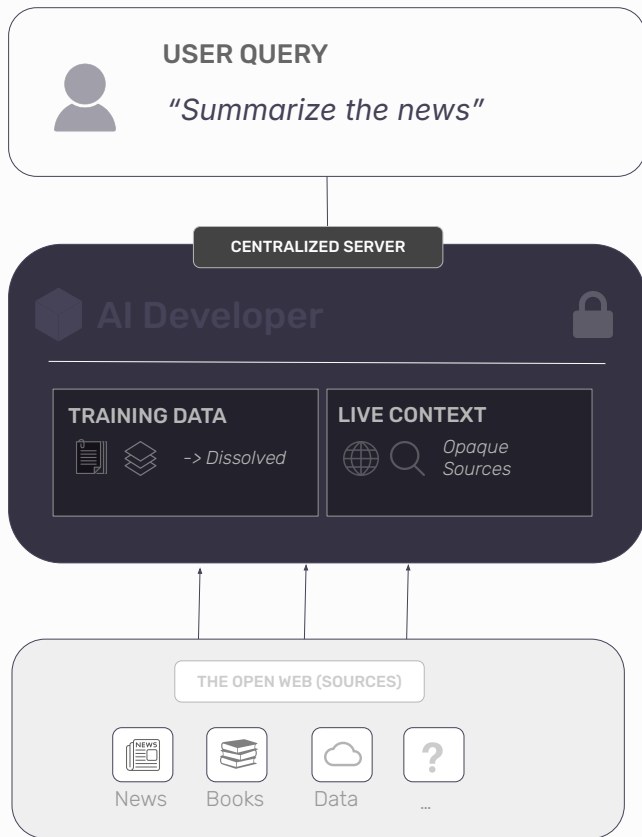


The ABC way

- The AI queries content live, at inference time
- Structured access to insights, not a copy of your files



NSAI for journalism

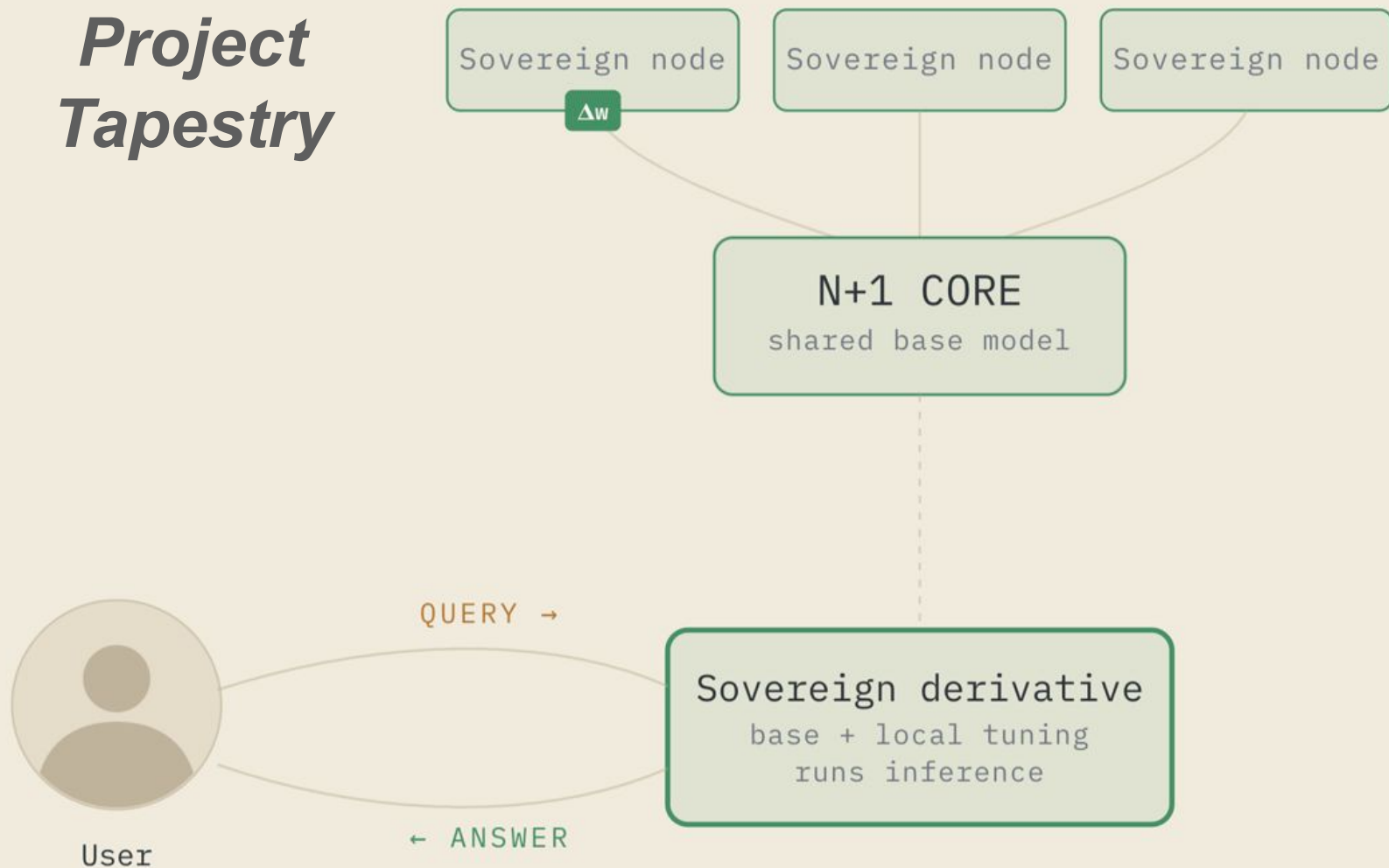


NSAI for sovereign, open, frontier models

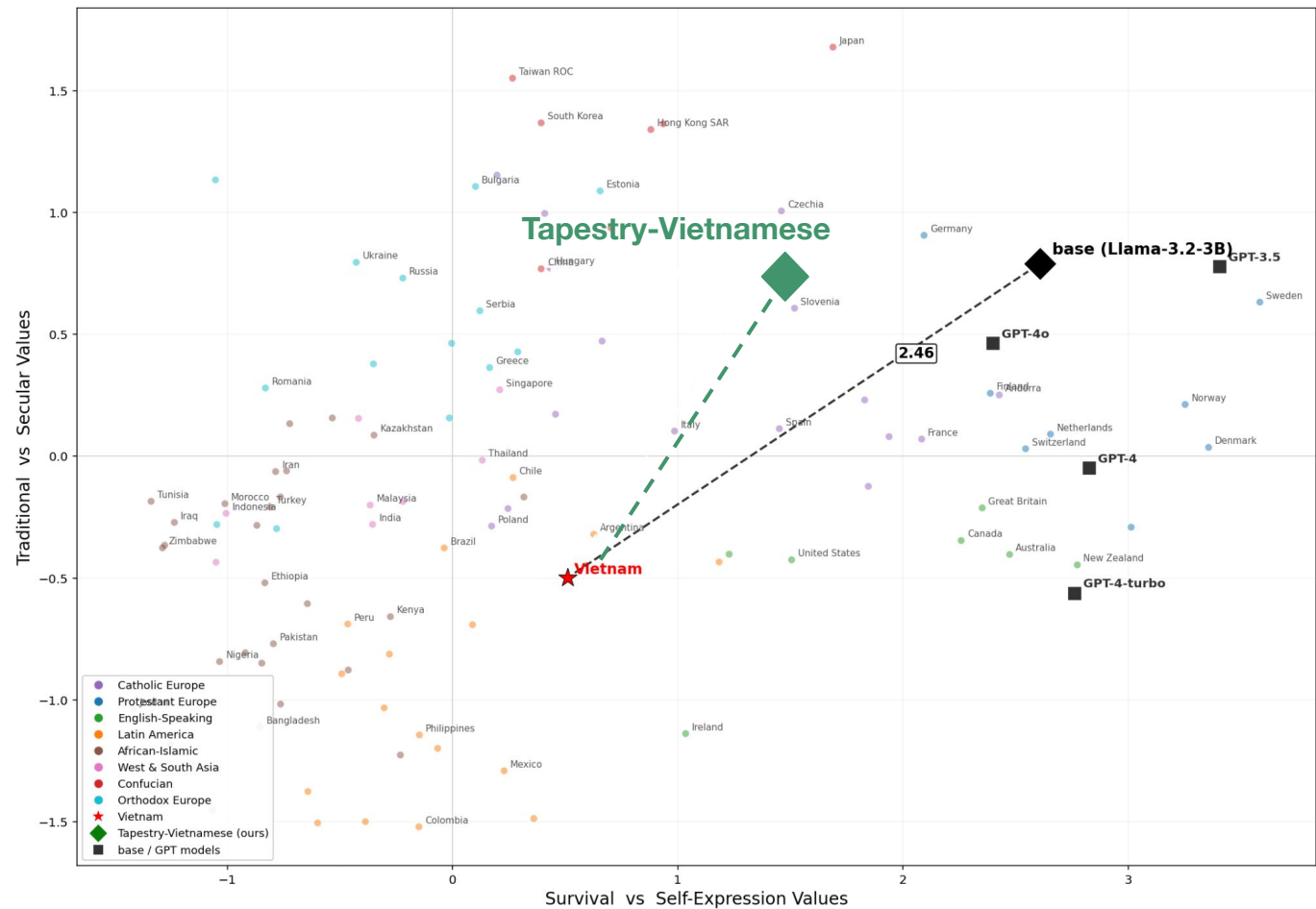
NSAI for sovereign, open, frontier models

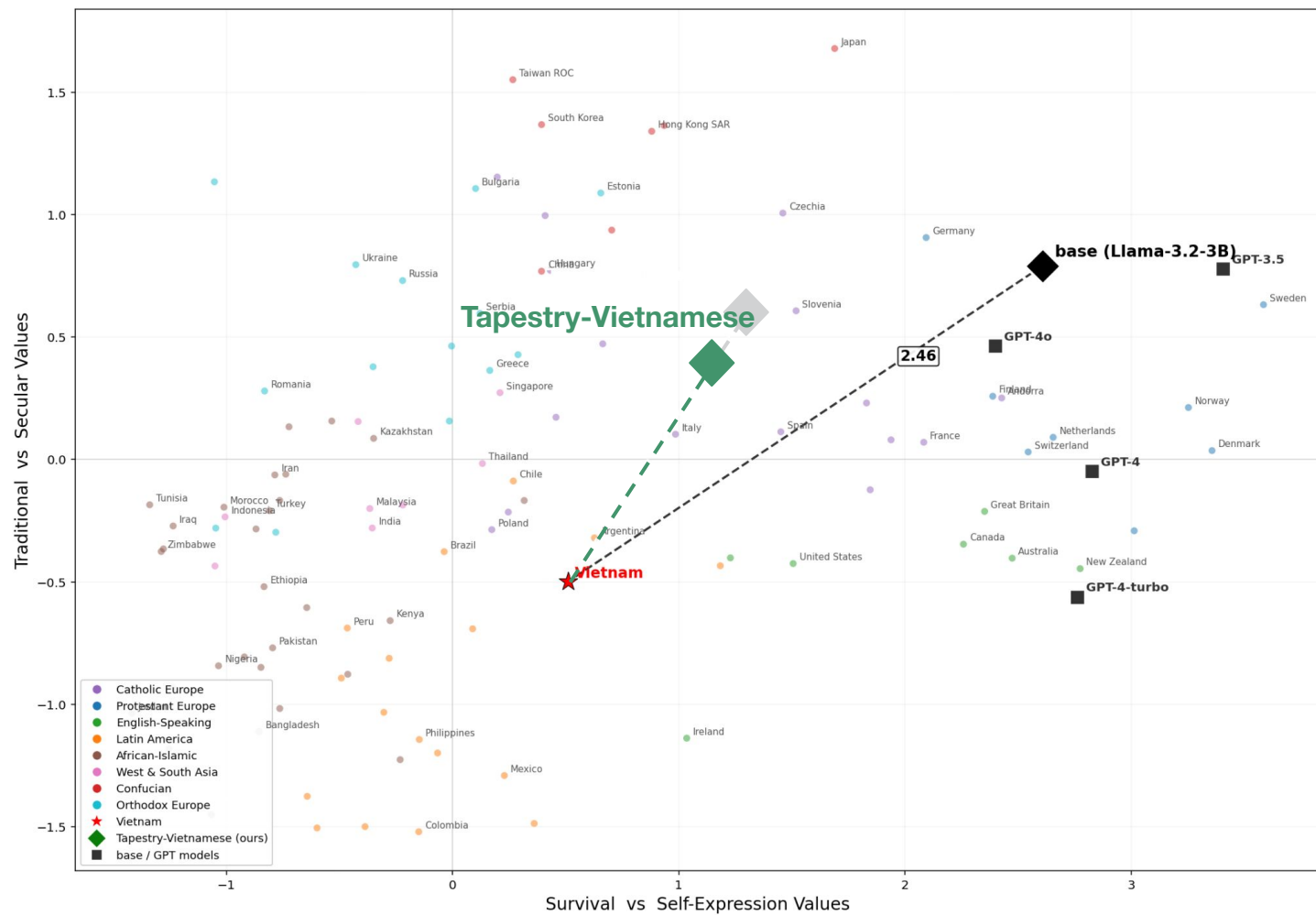
Project Tapestry

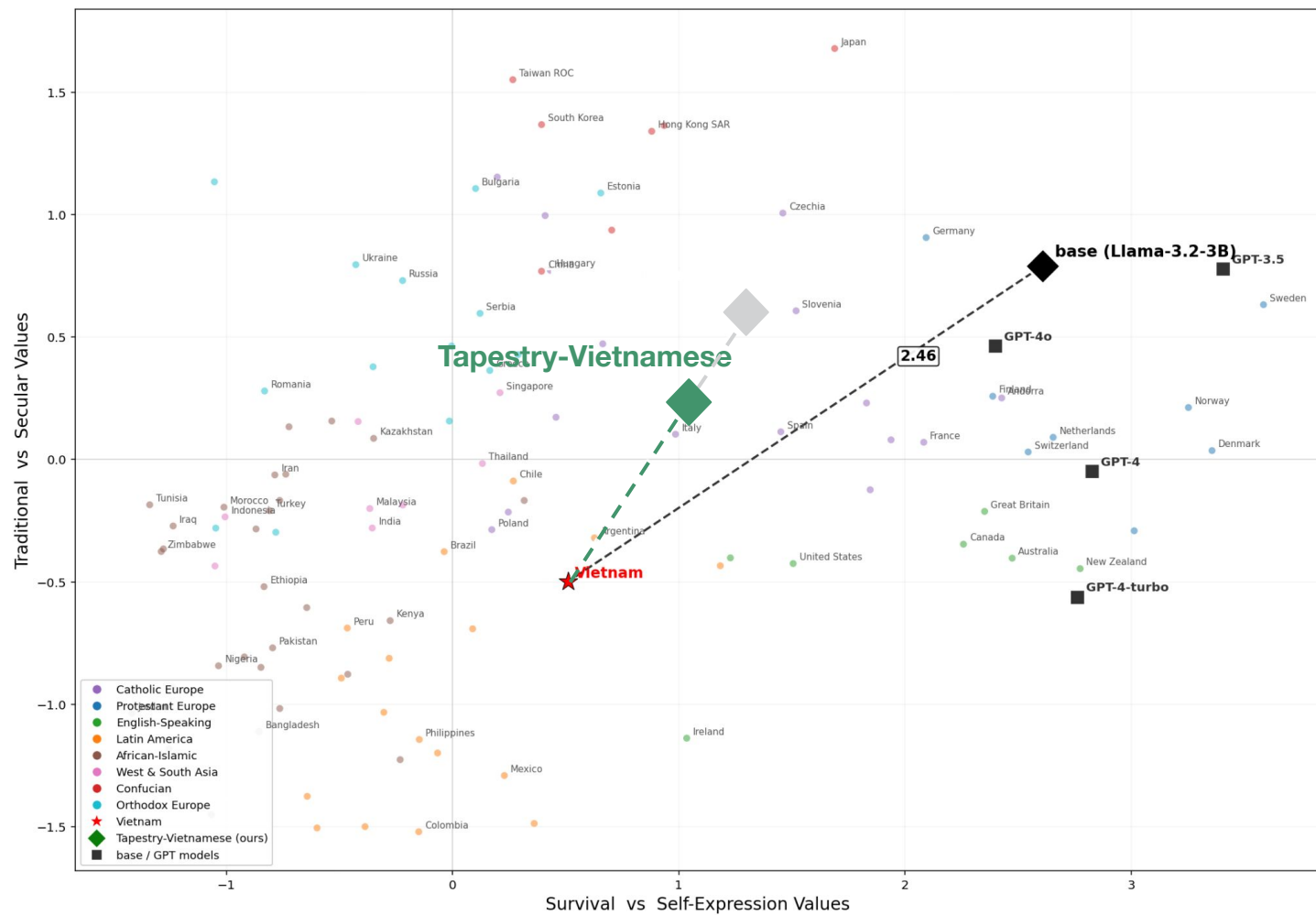
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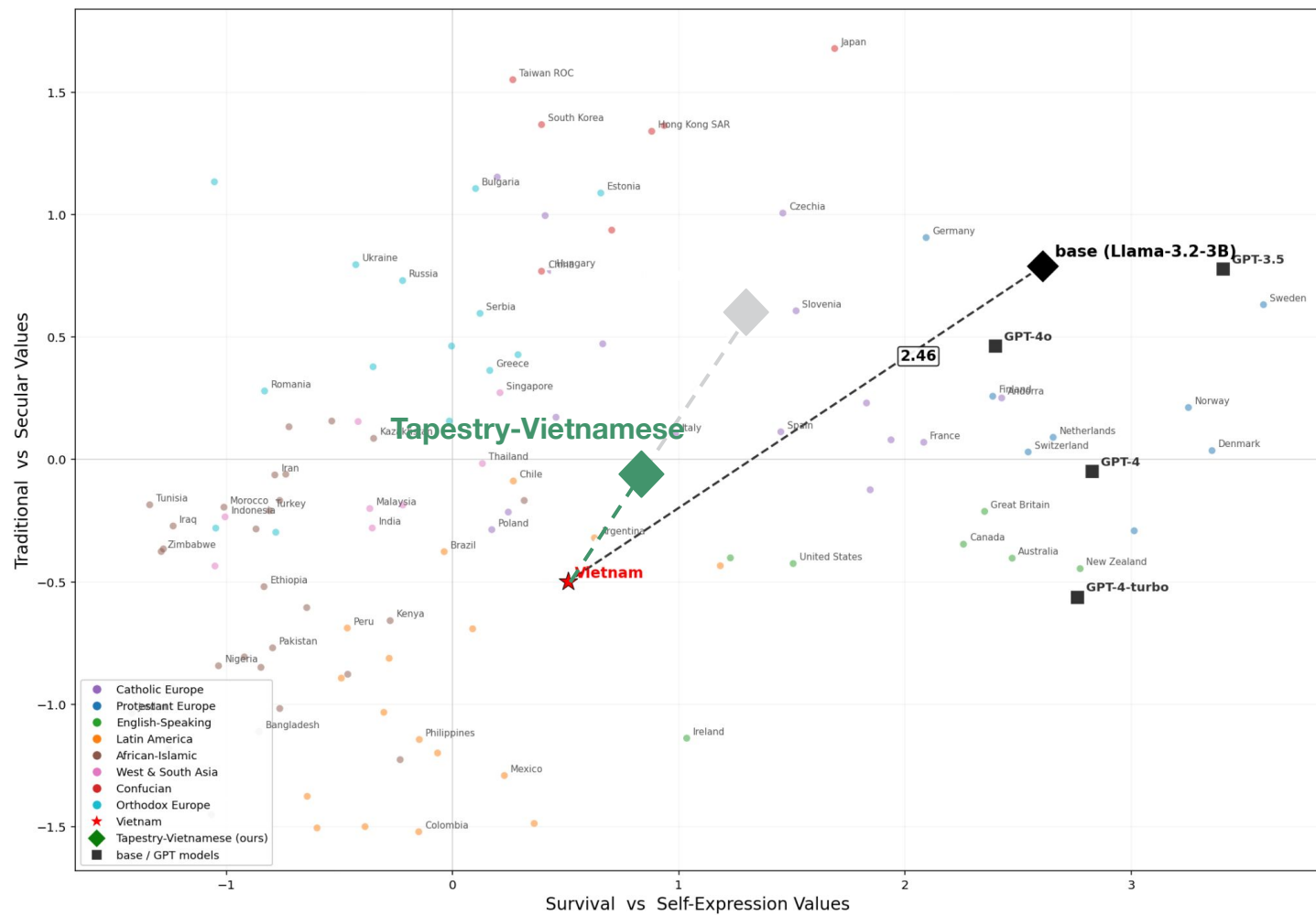


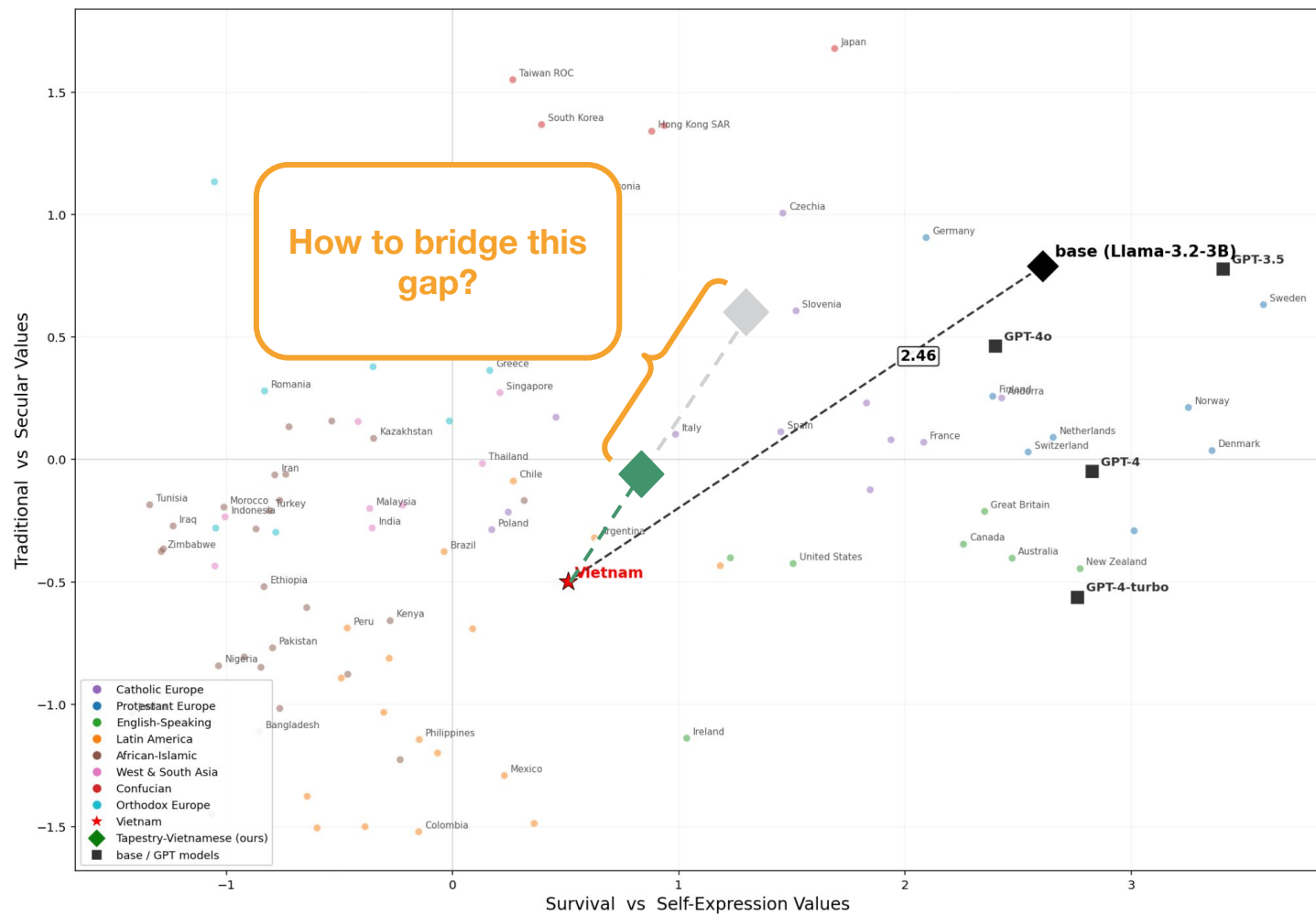
Tapestry-Vietnamese on the Inglehart-Welzel map



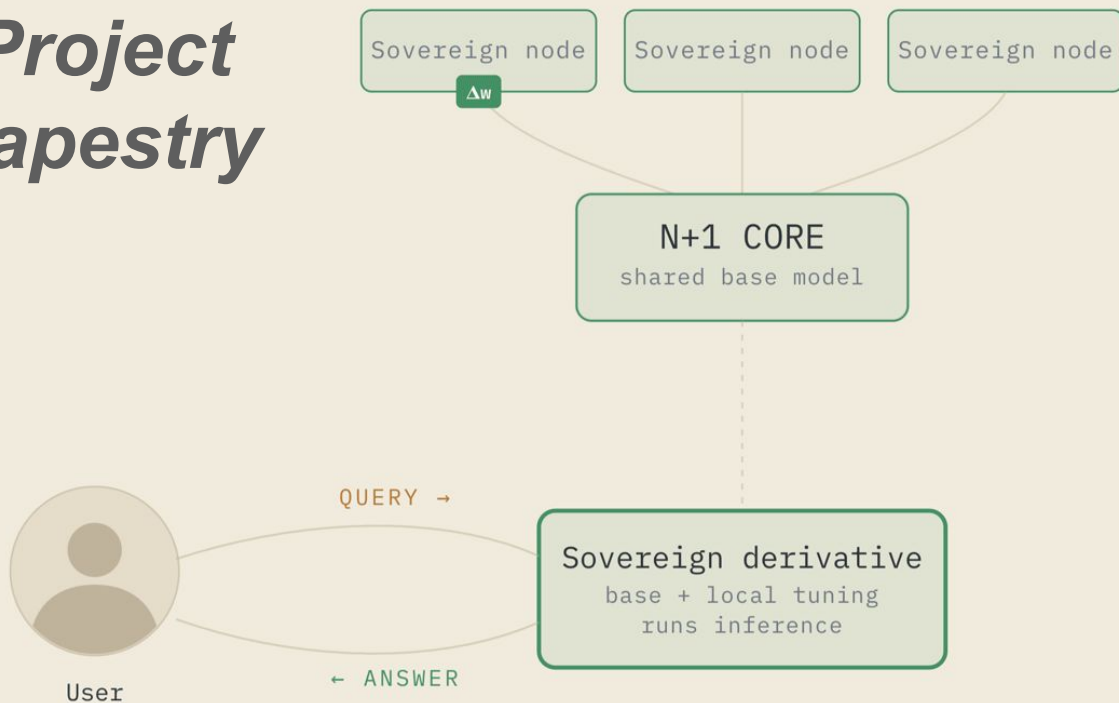




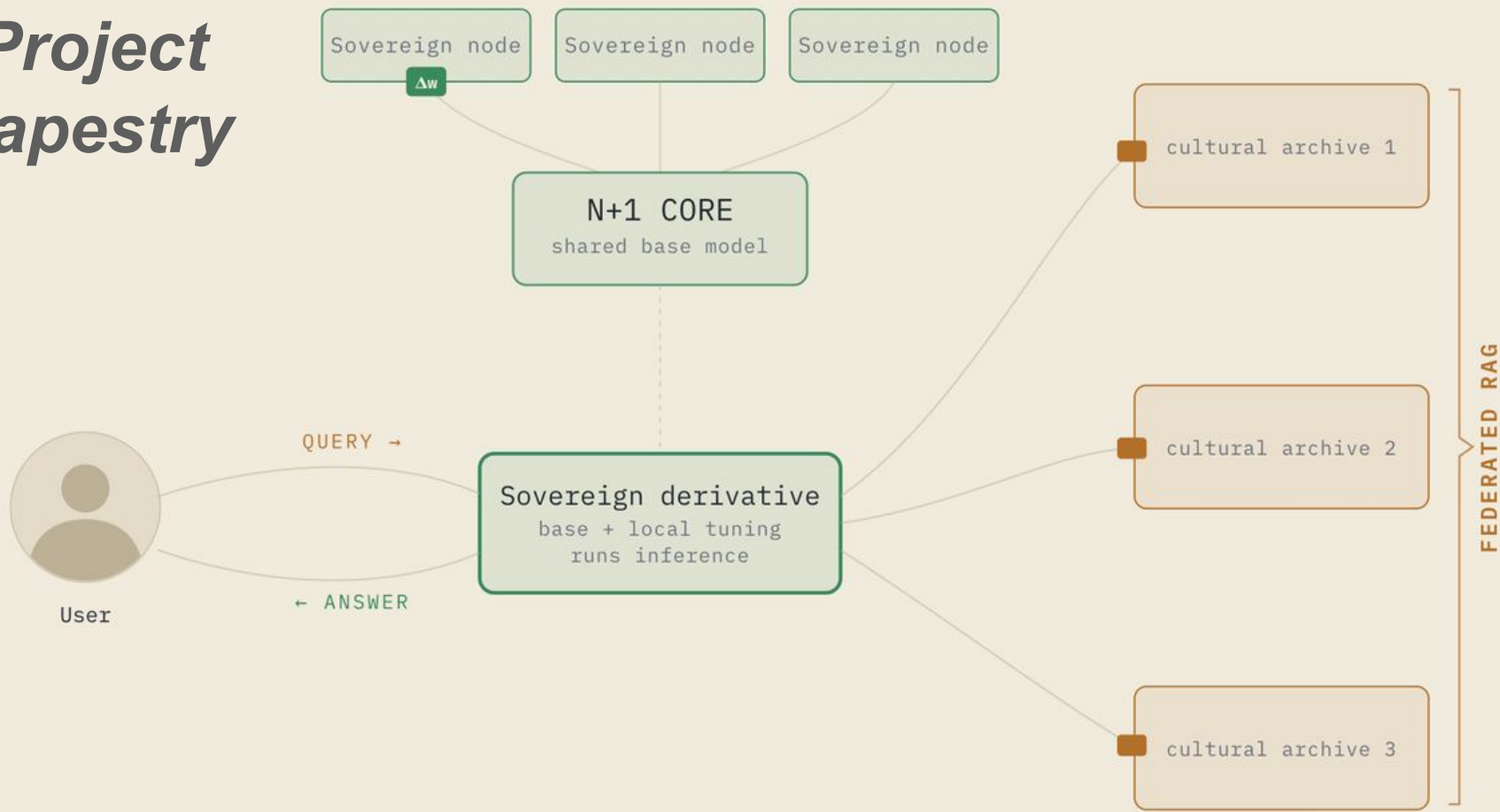




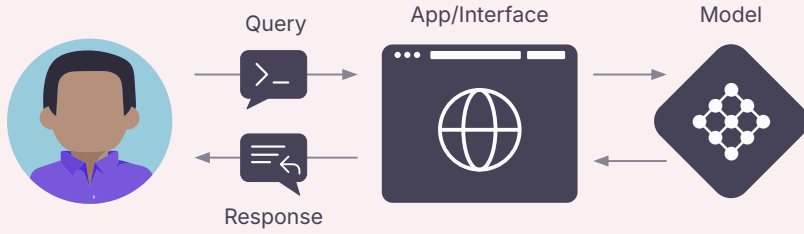
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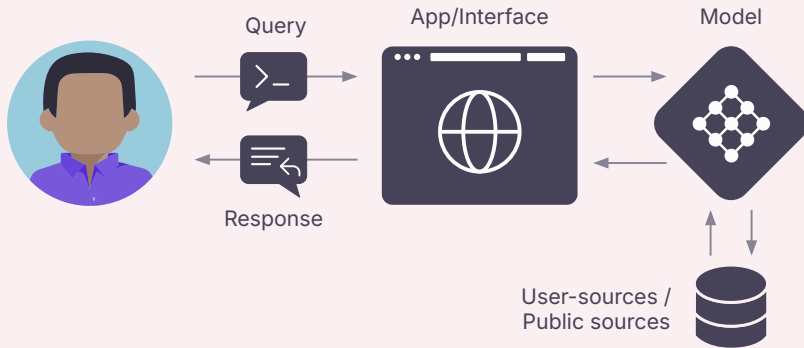
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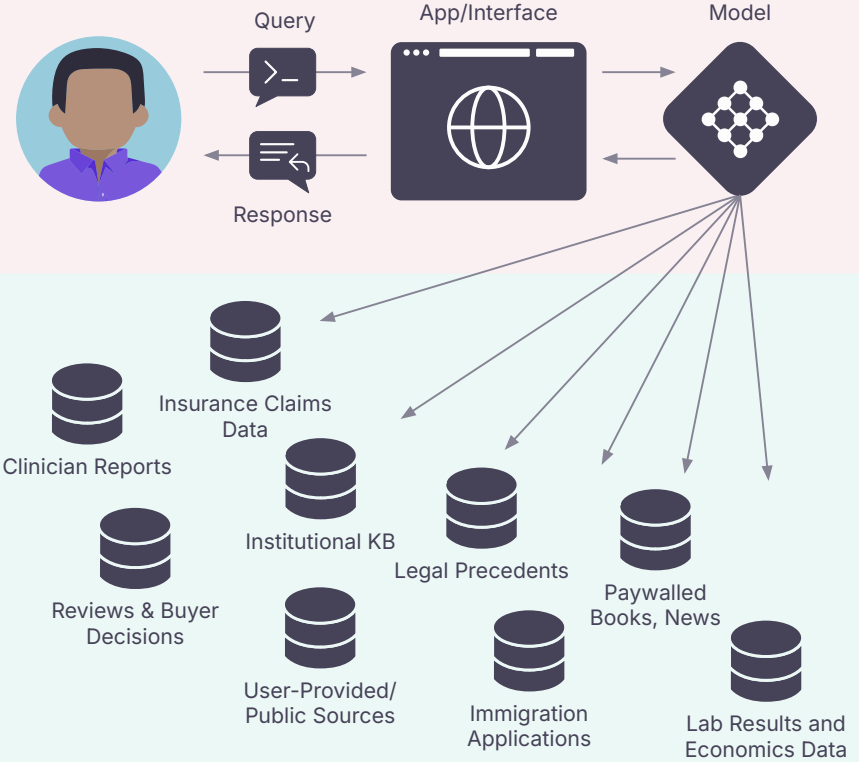
Large Language Model (LLM)

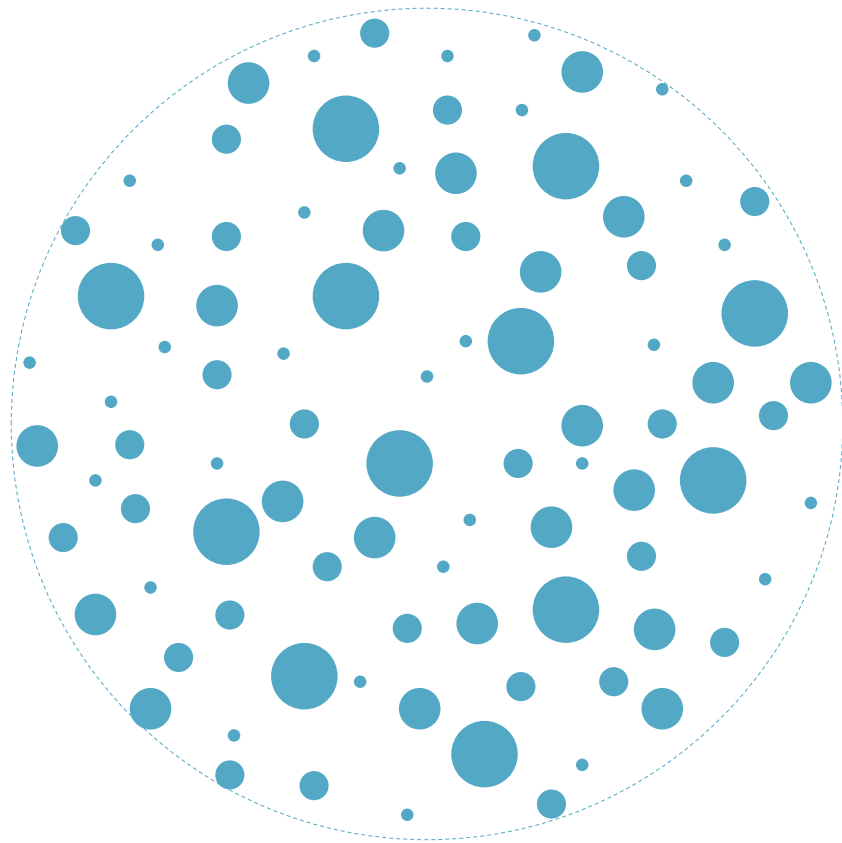


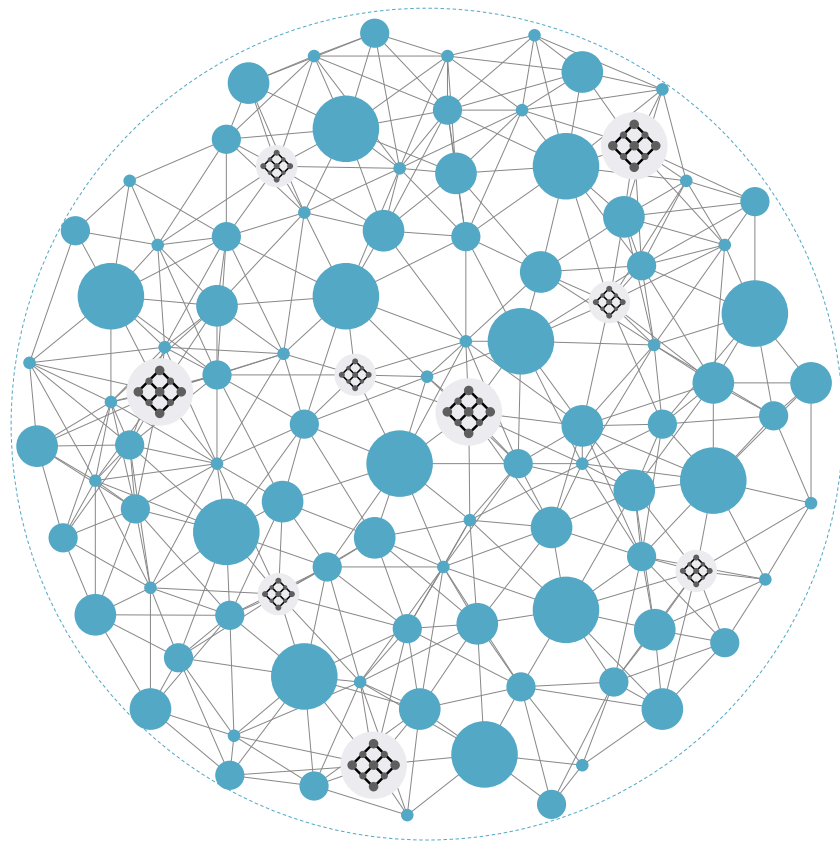
Retrieval Augmented Generation (RAG)

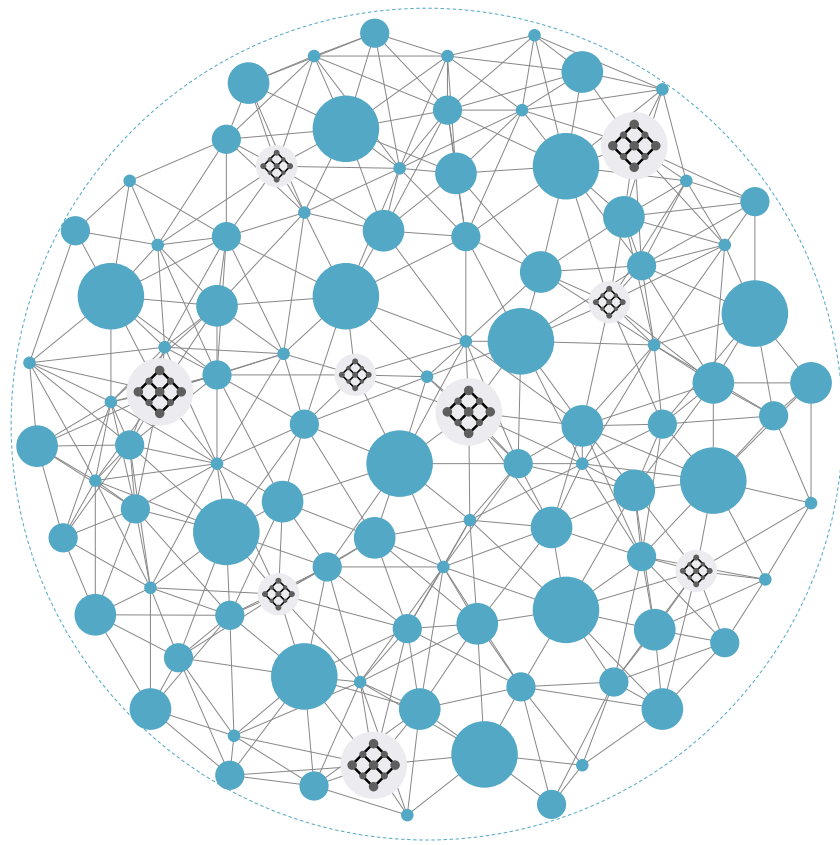


Federated Retrieval Augmented Generation (FedRAG)









***Sovereign models drawing on sovereign data =>
meaningfully sovereign AI***

Further reading

- syft.docs.openmined.org
- attribution-based-control.ai
- syfthub.openmined.org/hi/
- openmined.org/blog/network-sourced-ai/



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Thank you!